



L29-155HF(L)-Type4 Technical Data Sheet For Lenovo

SnBi Low temperature solder paste

Benefits of MILATERA L29-155HF(L)-Type4

TC-2305011J(L)
P2/20



01



Low temp. assembly

Reflow peak temp. can be reduced from 240°C to 170°C, which reduces CO₂ emissions

02



Improvement of drop shock reliability

Enhance ductility to improve drop shock reliability

03

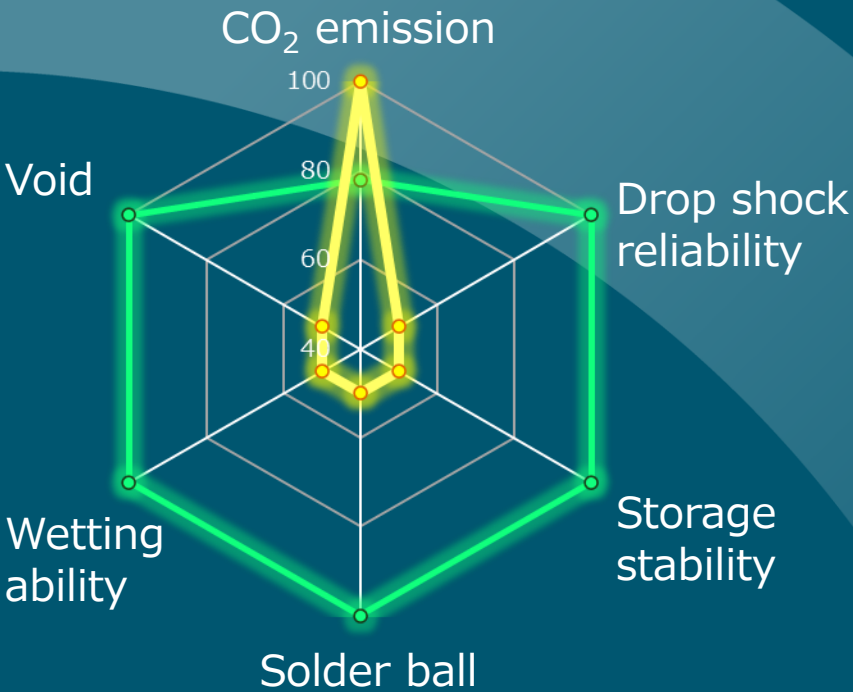


Expansion of the assembly process windows

Improve storage stability, stencil life and reflow ability

SnBi Low temperature solder paste

Ability chart of MILATERA L29-155HF(L)-Type4



SAC305 paste
Conventional Sn-58Bi paste

Chart 1

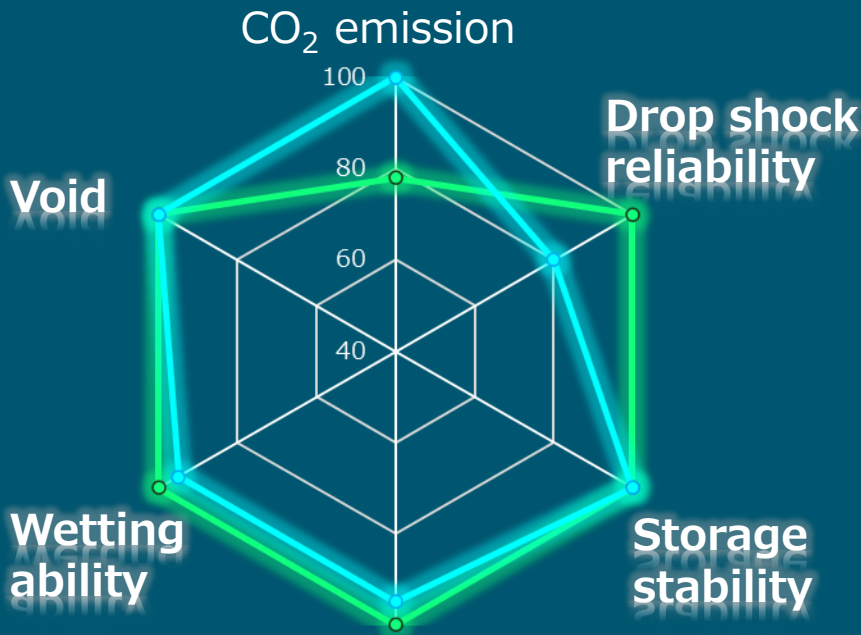
Comparison between
SAC305 & conventional SnBi paste

L29-155HF(L)-Type4

High
ductility
L29 alloy

155HF(L)
flux

Improve drop reliability,
storage stability
and reflow ability
(Solder ball · Wetting · Void)



Solder ball
SAC305 paste
L29-155HF-Type4

Chart 2

Comparison between
SAC305 & L29-155HF(L)-Type4

SnBi Low temperature solder paste

Effect of MILATERA L29-155HF(L)-Type4

TC-2305011J(L)
P4/20

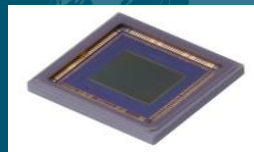


Raw material production
CO₂ emission
86% cut down

Production line
CO₂ emission
22% cut down

Ratio of product failure
Actual value
30% reduced

Adaptability
Weak heat resistance
component



Sensor



Organic EL

Multi component assembly



Large size insertion components
can be reflowed

Low temp. assembly

- Lower temp. setting of reflow oven to reduce CO₂ emissions ...P. 6

Improvement of drop shock reliability

- Development of L29 alloy improved ductility ...P. 7,8

Expansion of the assembly process windows

- Concept of 155HF(L) flux ...P. 9
- Storage stability ...P. 10
- Stencil life ...P. 11
- Reflow ability ...P. 12~14

Other properties

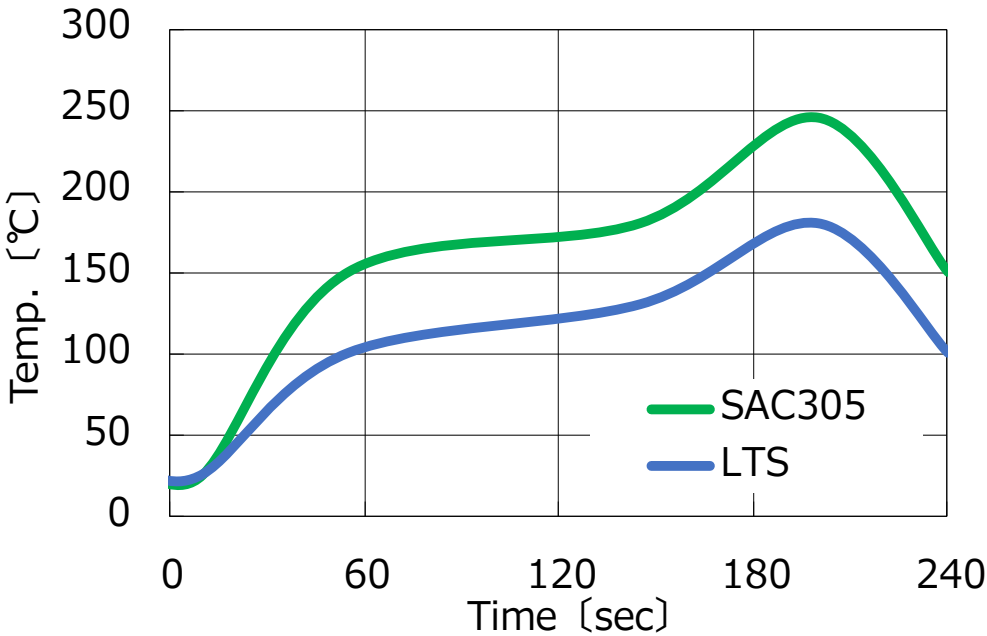
- Cu plate, Cu mirror corrosion test ...P. 15
- Migration test ...P. 16
- Handling guidelines ...P. 17,18
- Recommended reflow profile ...P. 19
- Data sheet ...P. 20

Low temp. assembly

Lower temp. setting of reflow oven to reduce CO₂ emissions



SNR-840GT



Contents	Conditions
Driving setting	Start up~8hrs/days 20days driving/months
CO ₂ emission factor	0.318kg-CO ₂ /kWh (2019 Record from Kansai Electric Power)

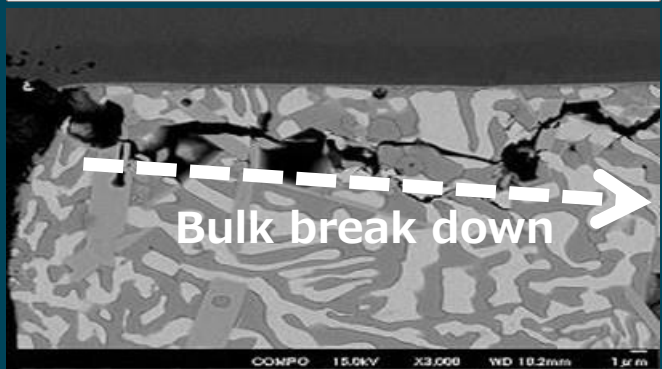
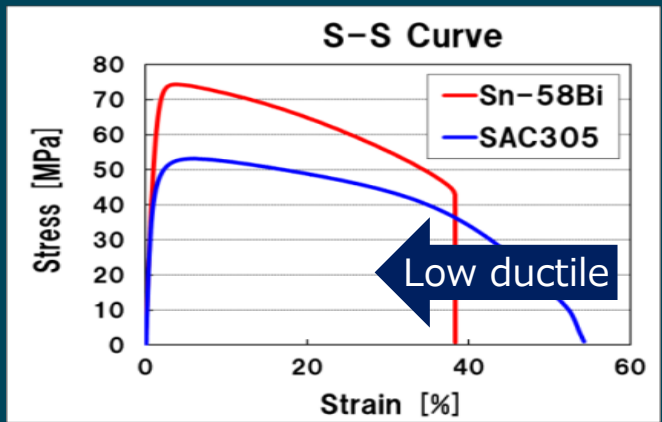
■ Comparison CO₂ emissions between SAC305 and Sn-Bi low temp. solder

Solder alloy	Reflow zone	Energy consumption (kWh)	Running time in one year (hrs)	Annual energy consumption (kWh/year)	Annual CO ₂ emission	CO ₂ cut down ratio (Comparison with M705)
M705 (SAC305)	8 zone	11.9	1920	22,848	7.27 t	-----
L29 (SnBi)	8 zone	9.25	1920	17,760	5.65 t	22 %

Improvement of drop reliability

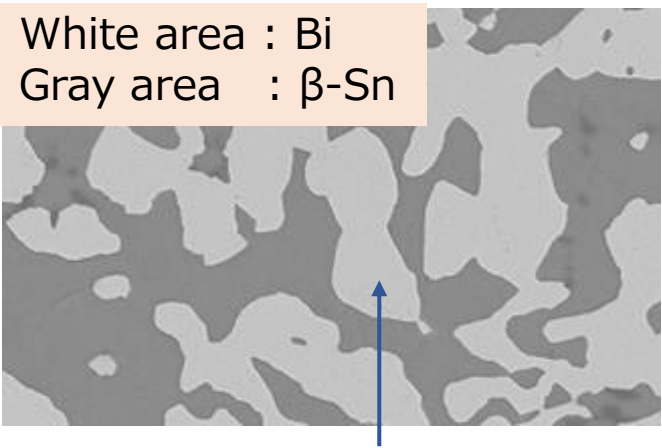
Development of L29 alloy improved ductility

Bi is hard and brittle



Hard alloy
⇒ High TCT reliability,
but weak drop reliability

L20(Sn-58Bi)



L29(Sn-58Bi-Sb-Ni)



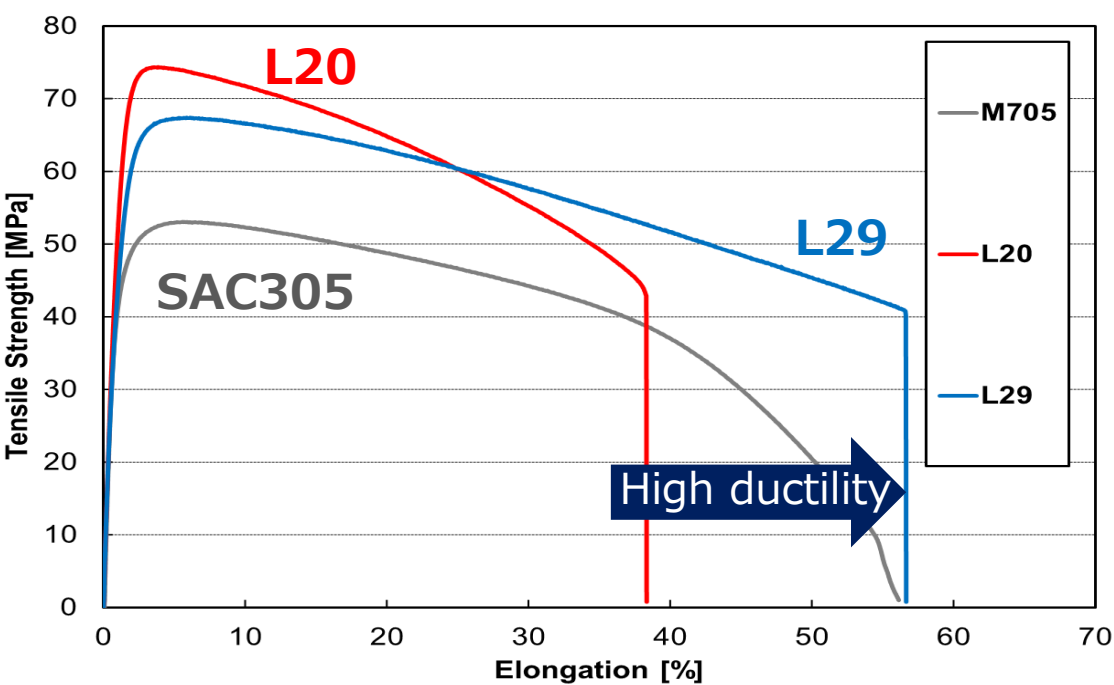
	Composition	Melting Temperature	Tensile Strength[MPa]	Elongation [%]
L20	Sn-58Bi	139 - 141	74.2	37
L23	Sn-57Bi-1.0Ag	138 - 204 *	74.3	41
L28	Sn-35Bi-0.5Cu-Ni	141 - 182	81.3	54
L29	Sn-58Bi-0.5Sb-Ni	140 - 145	68.6	58
M705	Sn-3.0Ag-0.5Cu	217 - 220	53.3	56

*204℃ is minute peak at DSC, therefore L23 can reflow at 170℃ peak.

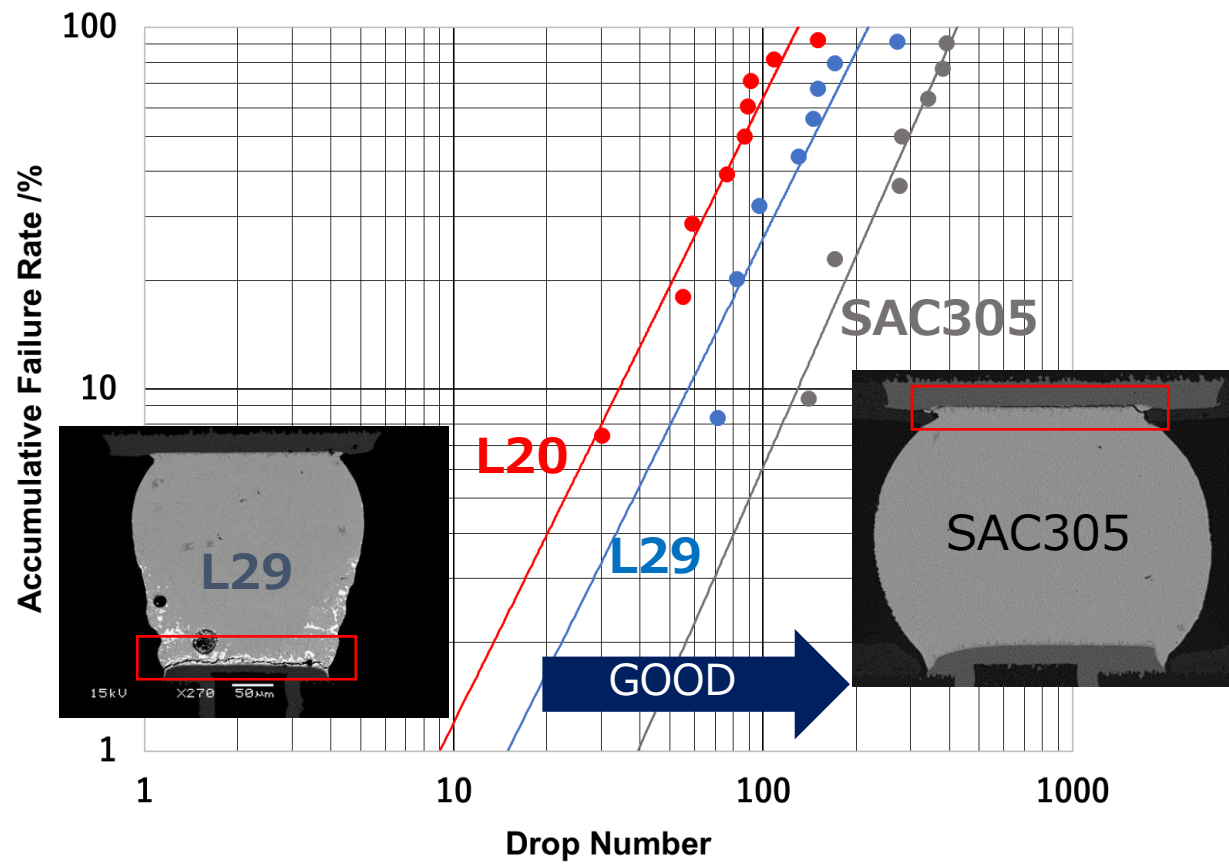
Improvement of drop reliability

Development of L29 alloy improved ductility

■ Stress-Strain curve



■ Drop shock reliability(1500G)



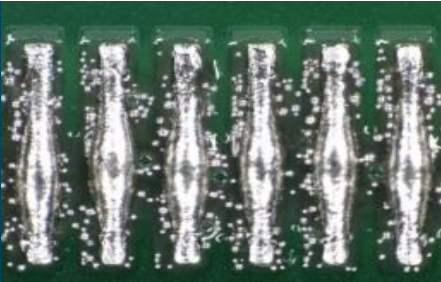
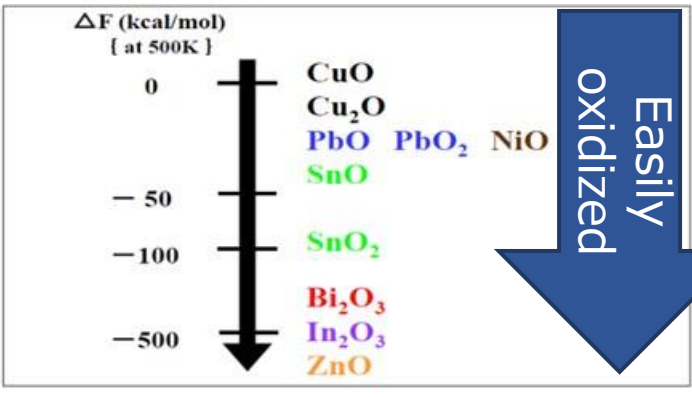
L29 has higher ductility and higher drop shock reliability than Sn-58Bi solder.

Expansion of the assembly process windows

Concept of 155HF(L) flux

Bi is oxidized easily

Standard generated Gibbs energy of metal oxide



Strong reduction power
to achieve good reflow ability



Flux reactions with
solder powder too much

■ New technology used for 155HF(L)

Improve reflow ability

Inhibit oxidation

Bi oxide

Achieve inhibition of Bi oxidation

A diagram illustrating the inhibition of Bi oxidation. It shows Bi atoms (blue circles) and Bi oxide (Bi atoms bonded to O atoms, orange circles). A red 'X' is placed over the Bi oxide formation, indicating inhibition. A white arrow points from the Bi atoms to the Bi oxide, suggesting the process being inhibited.

A microscopic image showing several solder joints on a green PCB. The joints are clean and well-defined, indicating improved reflow ability.

Less solder ball

Improve shelf life (Stencil life)

Inhibit powder agglomeration

A diagram illustrating the inhibition of powder agglomeration. It shows Bi atoms (blue circles) and Bi oxide (Bi atoms bonded to O atoms, orange circles). A red 'X' is placed over the Bi oxide formation, indicating inhibition. A white arrow points from the Bi atoms to the Bi oxide, suggesting the process being inhibited.

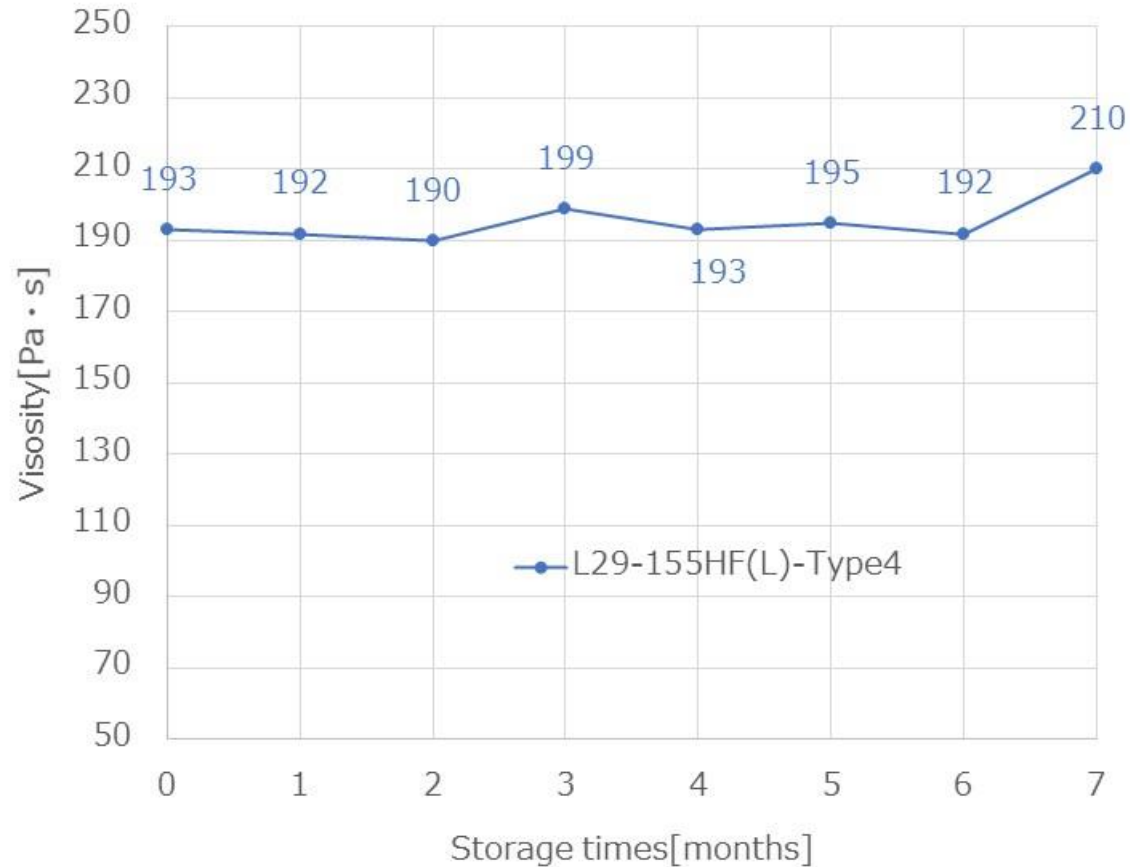
A microscopic image showing a single solder joint on a green PCB. The joint is clean and well-defined, indicating improved printing.

Improve printing defect

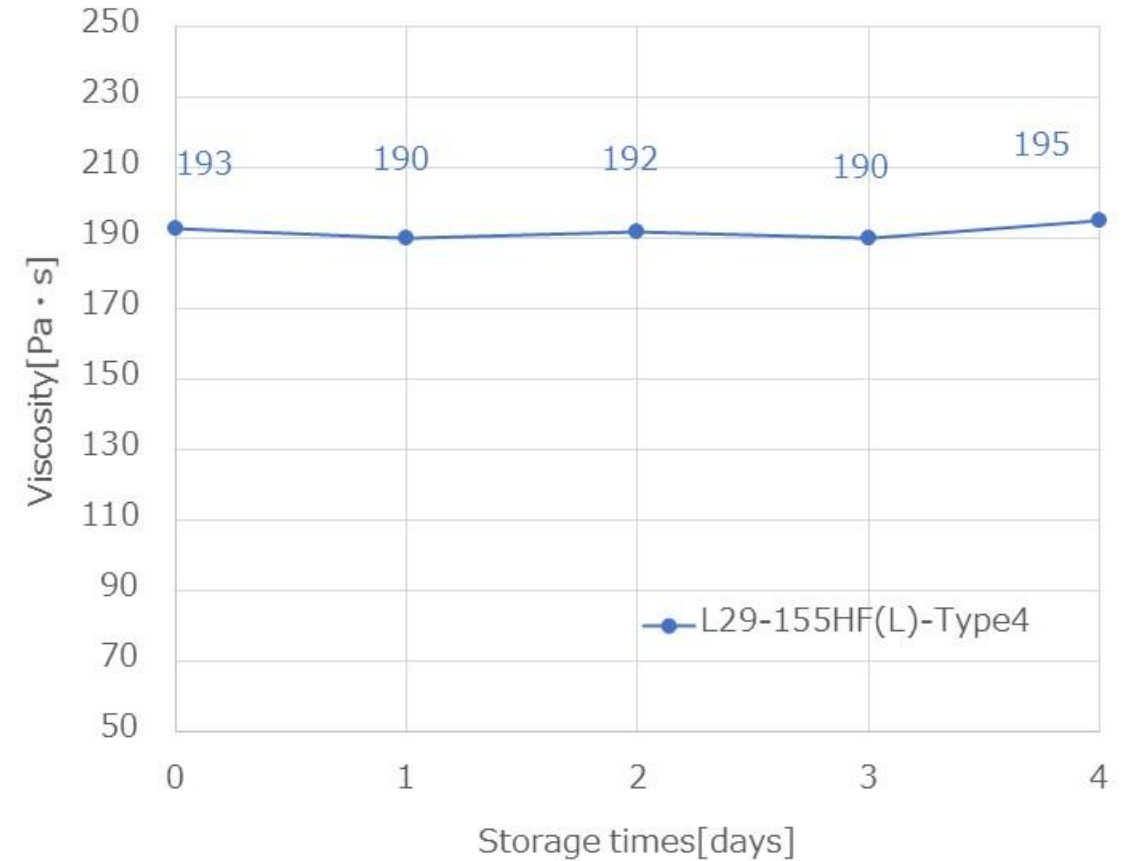
Expansion of the assembly process windows

Storage stability of L29-155HF(L)-Type4

■ Viscosity change in storage (Refrigeration)



■ Viscosity change(Room temp.)

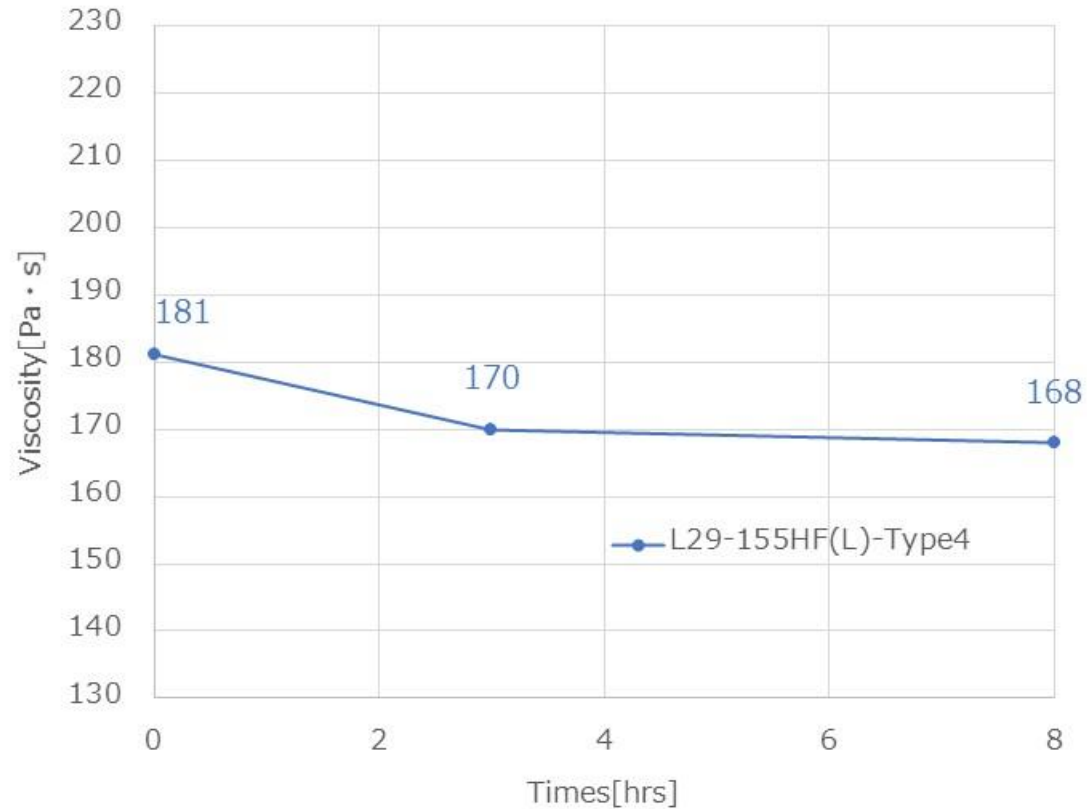


Viscosity of 155HF(L)-Type4 didn't increase for 6 months in refrigerator and for 4 days in room temperature.

Expansion of the assembly process windows

Stencil life of L29-155HF(L)-Type4

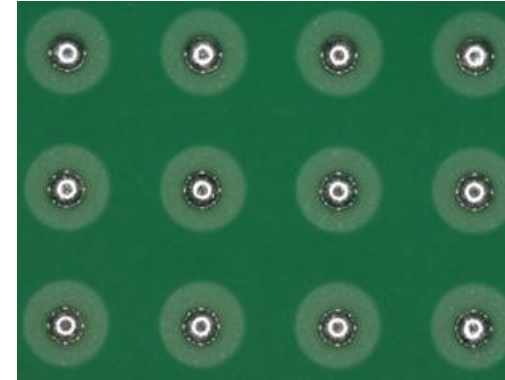
■ Viscosity change in continuous printing



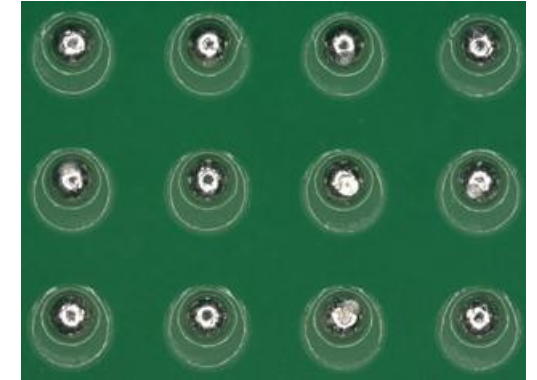
Viscosity of 155HF(L)-Type4 didn't increase after 8hrs continuous printing.

■ Reflow ability

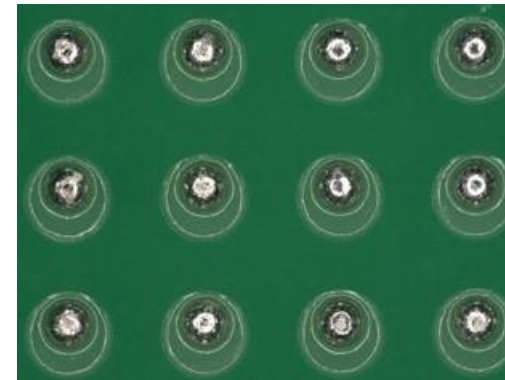
Stencil thickness: 0.15mm Pad size: 0.30mm
PCB surface : Cu-OSP



0hr



3hrs



8hrs

Reflow ability of 155HF(L)-Type4 didn't change after 8hrs continuous printing.

Expansion of the assembly process windows

Reflow ability of L29-155HF(L)-Type4

■ Leaving test after printing(28°C60%RH)

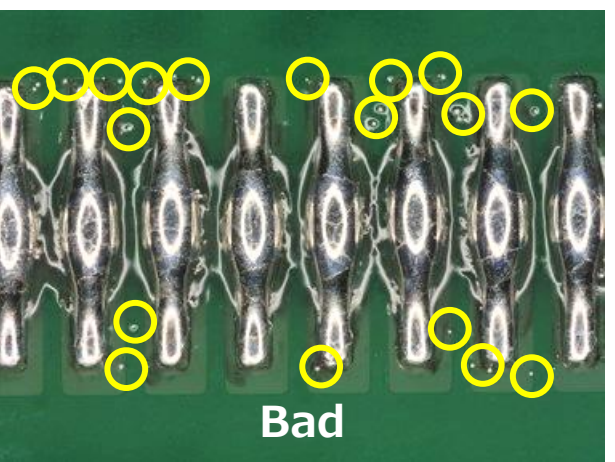
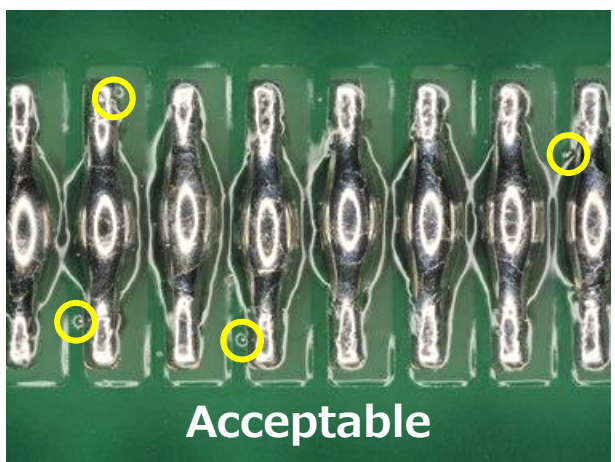
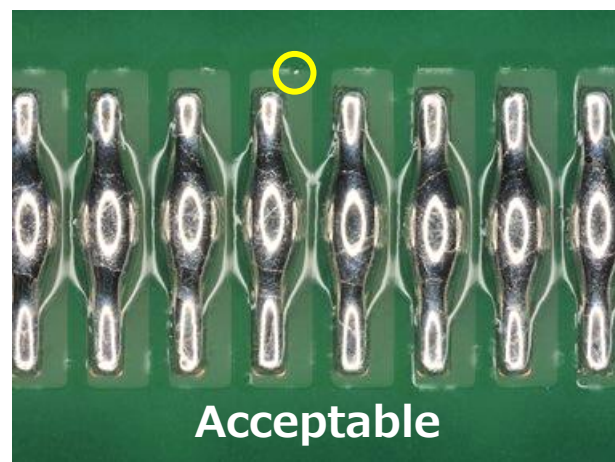
Stencil thickness: 0.15mm Pad size: 0.30mm
PCB surface : Cu-OSP
4hrs

Initial

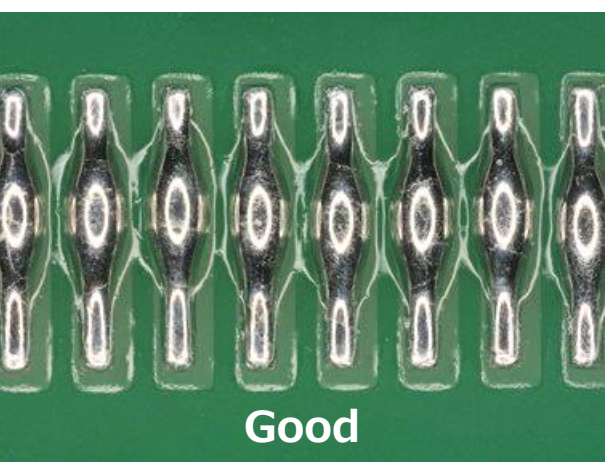
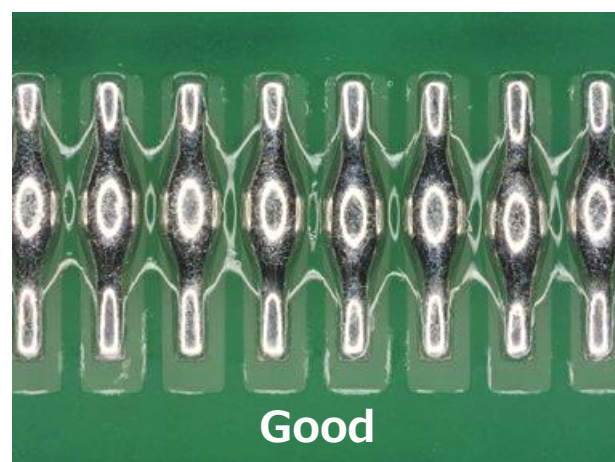
2hrs

4hrs

Sn-58Bi
conventional
paste



155HF(L)



155HF(L) can be left for up to 4hrs after printing.

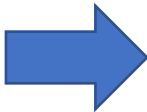
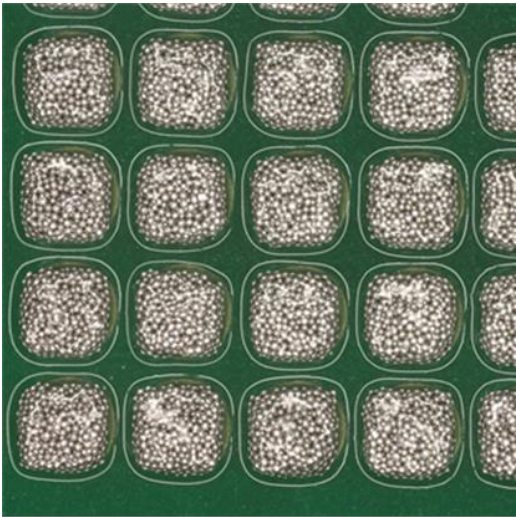
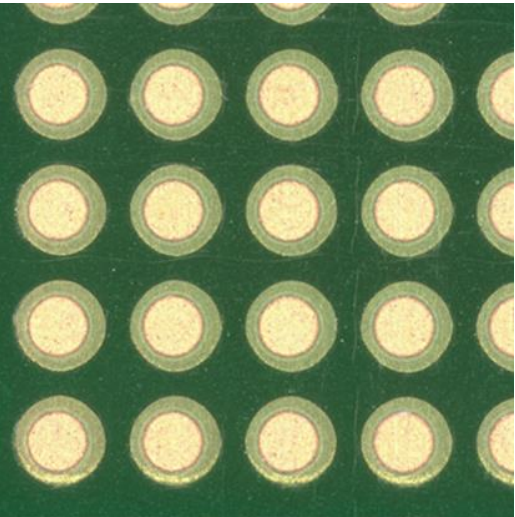
Expansion of the assembly process windows

Reflow ability of L29-155HF(L)-Type4

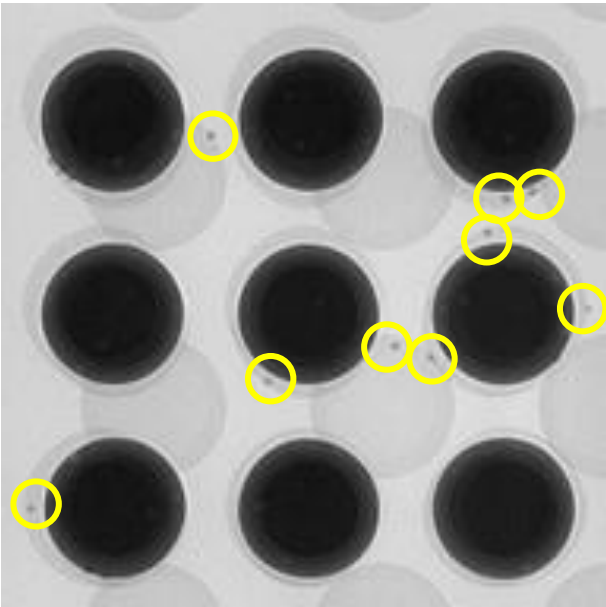
■ Solder ball

Before printing

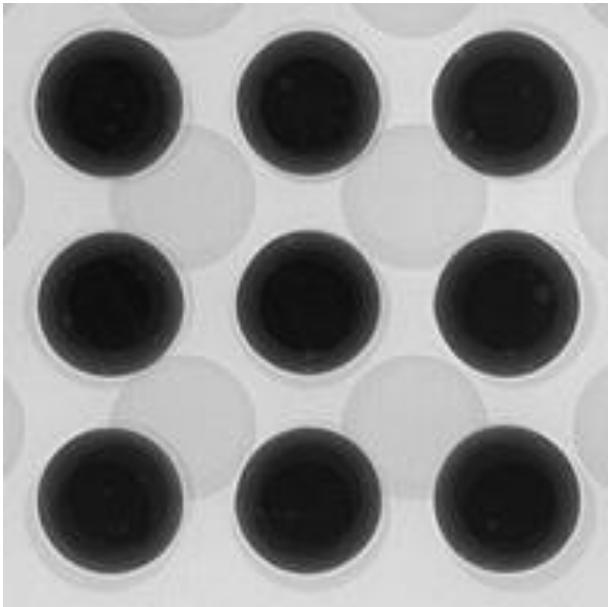
After printing



Sn-58Bi
conventional paste



155HF(L)



PCB surface : Cu-OSP
Pad size : $\Phi 0.26\text{mm}$

Stencil thickness : 0.10mm
Aperture size : $\square 0.38\text{mm}$

Printing aperture was larger than Pad.

BGA size : $\square 15\text{mm}$
BGA ball component : SAC305
BGA ball size : $\Phi 0.3\text{mm}$

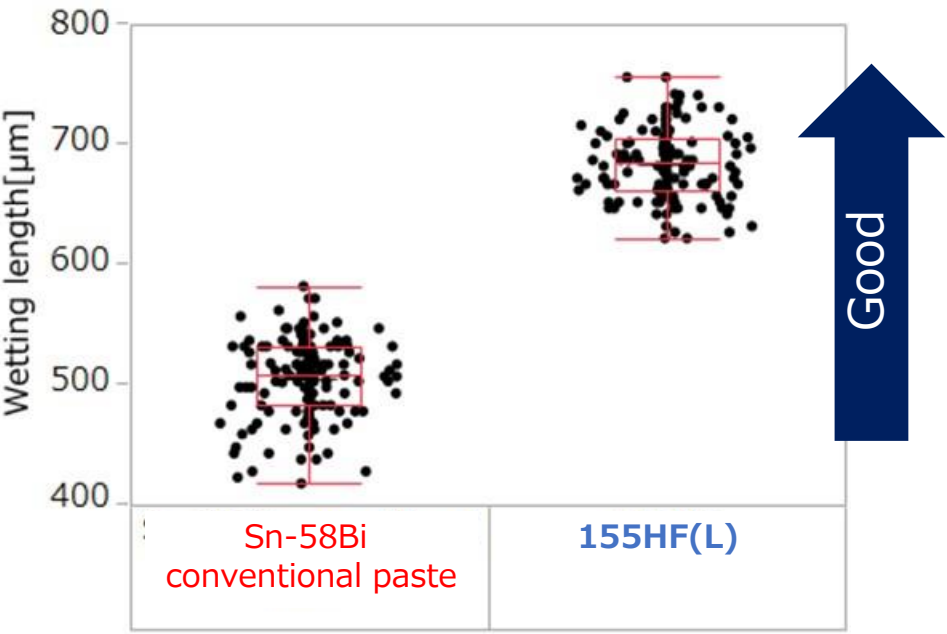
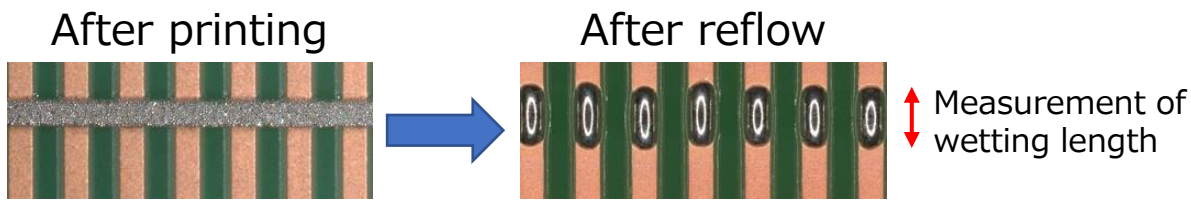
Solder ball under BGA was improved.

Expansion of the assembly process windows

Reflow ability of L29-155HF(L)-Type4

Wetting ability

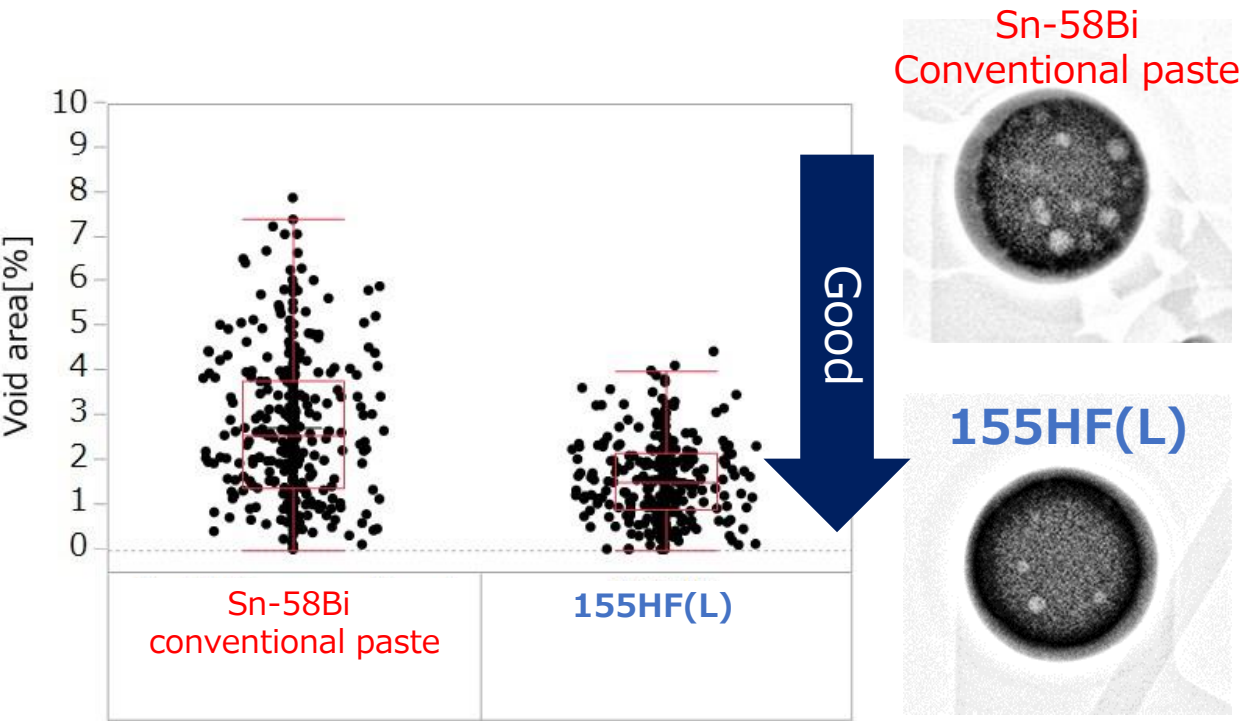
Stencil thickness : 0.15mm
PCB surface : Cu-OSP



155HF(L) has good wetting ability for Cu.

Void

Stencil thickness: : 0.12mm
PCB surface : Cu-OSP
BGA ball composition : SAC305
Pad size : 0.46mm
BGA size : □15mm
BGA ball size : Φ0.4mm



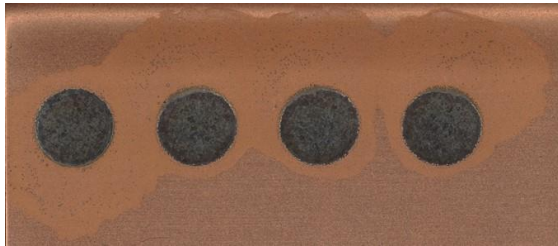
Void area under BGA of 155HF(L) was less than 5%.

Other properties

Cu plate, Cu mirror corrosion test

■ Cu plate corrosion test

Procedure : JIS Z 3197
Environment : 40°C90%RH
Time : 72hrs



155HF(L)

blank

Results: Pass

■ Cu mirror corrosion test

Procedure : JIS Z 3197
Control material : WW rosin 25% IPA solution
Test material : 155HF(L) flux 25%IPA solution
Environment : 25°C50%RH
Time : 24hrs



155HF(L) solution

WW rosin solution

Results: Pass
Mirror breakthrough was not observed.

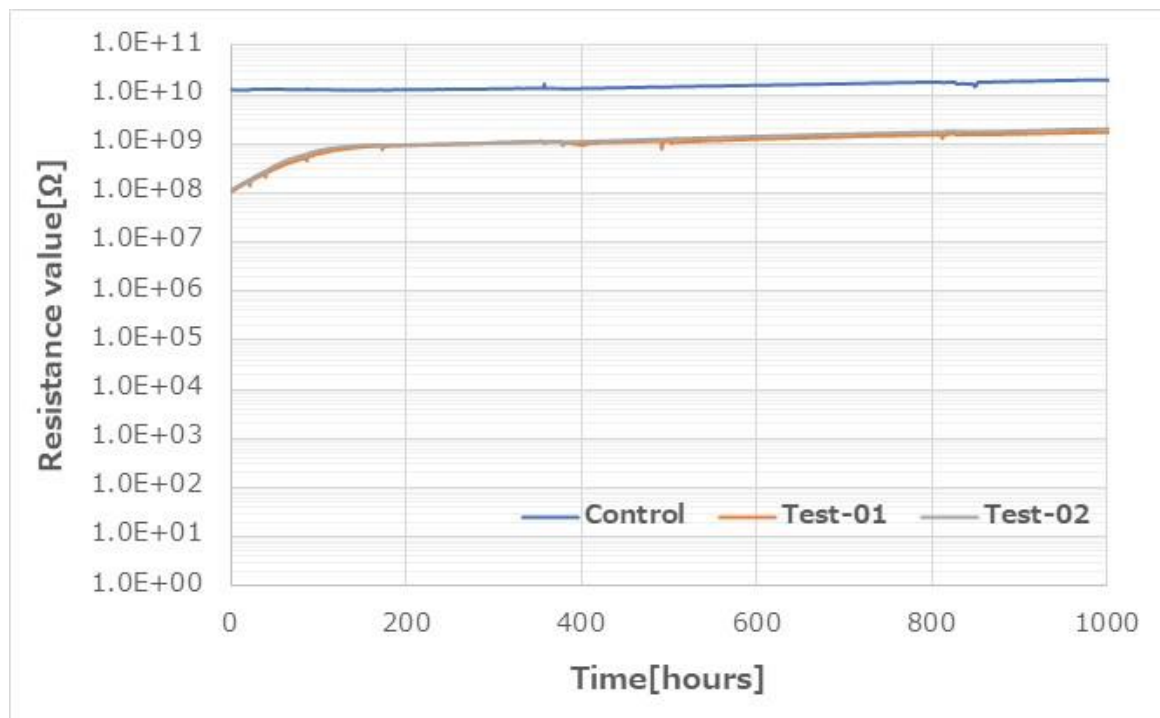
Other properties

Migration test

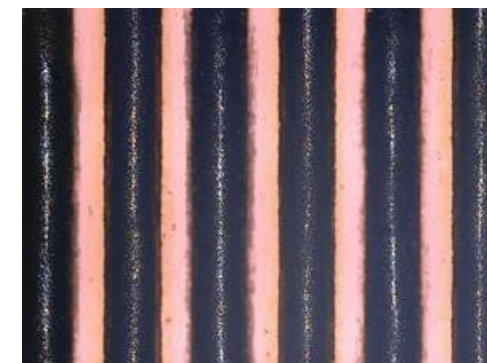
TC-2305011J(L)
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■ Migration test

- Line/Space : 0.165mm / 0.165mm
- Stencil thickness : 0.10mm
- Apply voltage : 50V
- Measurement voltage : 100V
- Test environment : 85°C/85%RH
- Criteria : Over $5.0 \times 10^8 \Omega$ (400, 600, 1000hrs)



Test-01 1000h



Test-02 1000h

Results: Pass
Migration was not observed.

Other properties

Handling guidelines

TC-2305011J(L)
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■ Recommended storage and handling

Item	Recommended procedure	Remark
Storage	6month at refrigerated 0 ~ 10℃	Without unsealing condition
Warm-up prior to use	At least 1hour leaving at ambient working temp	Do not use forced heating methods. Actual paste temp should be verified
Capable storage period at room temperature prior to use	4 days	Without unsealing condition below 25℃
Stirring before processing	Manual: 30 ~ 60sec Automatic: 30 ~ 60sec*	*Depend on stir rate of stirring machine (Caution against surplus stir)
Working environment	Temperature: 22 ~ 28℃ Related humidity: 20 ~ 60%RH	— — —
Capable working time in continuous printing	8 hrs	
Capable abandon time in printing process	Within 1 hr	— — —
Capable idle time before reflow after print	Within 4 hrs	— — —
Re-storage of remaining paste in container (unused portion)	At refrigerated 0 ~ 10℃	Allowed to re-storage 1time in validity with putting the lid on firmly.

Other properties

Handling guidelines

TC-2305011J(L)
P18/20

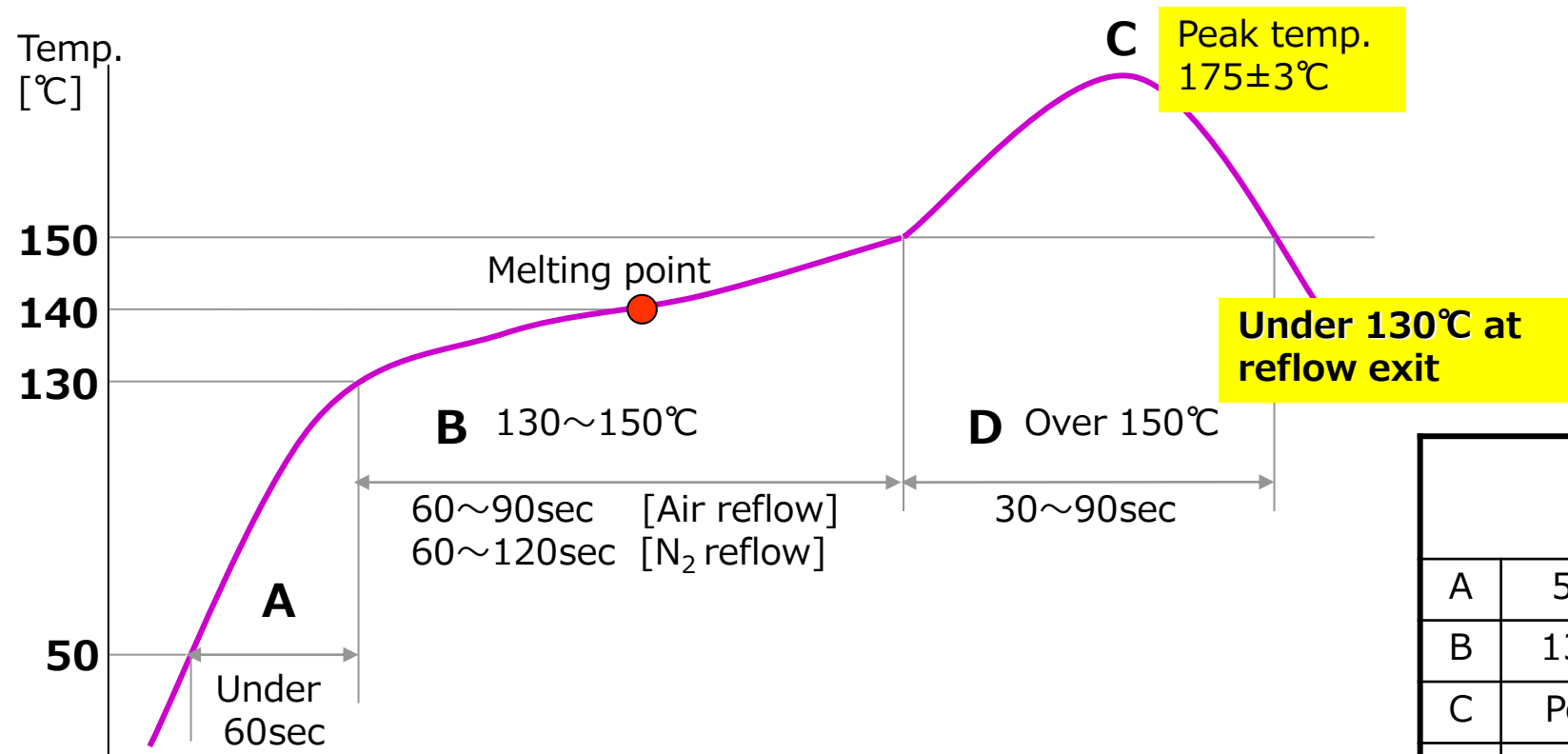
■ Recommended printing conditions

Item	Recommended procedure	Capable
Printer types	Open squeegee model	Enclosed squeegee system
Squeegee blade	Metal	Urethane, Plastic
Squeegee angle	60°	— — —
Printing speed	30 ~ 50mm/s	20 ~ 180mm/s
Printing pressure	0.20 ~ 0.30N/mm	Adjust not to leave paste on stencil
Separation speed	1.0 ~ 5.0mm/s	< 10mm/s
Paste rolling diameter	10 ~ 15mm	— — —
Temperature & Humidity	22 ~ 28℃ 20 ~ 60%RH	— — —

Other properties

Recommended reflow profile

Alloy : L29
Flux : 155HF(L)



*155HF(L)-Type4 is formulated for using in AIR reflow process.
But it is recommended using Nitrogen reflow
to obtain higher soldering quality and stability.

		Recommended Value	
		Air Reflow	Nitrogen Reflow
A	50~130°C	under 60sec	under 60sec
B	130~150°C	60~90sec	60~120sec
C	Peak temp.	175±3°C	175±3°C
D	Over 150°C	30~90sec	30~90sec
E	O ₂ Concentration	— — — —	under 1000ppm

Other properties

Data sheet

TC-2305011J(L)
P20/20

Items	L28-143HF(L) Type4	L29-155HF(L)-Type4	Test method
SOLDER POWDER			
Alloy Composition	L28 (Sn-35Bi-0.5Cu-Ni)	L29 (Sn-58Bi-0.5Sb-Ni)	- - -
Melting Temperature	141-182℃	140-145℃	DSC
Powder Shape	Spherical		SEM
Particle Size	Type4 (20-38μm)		SEM, Screen Method
SOLDER PASTE			
Flux Classification	ROL0		J-STD-004B
Halogen Content	Less than 900ppm		EN 14582
Halide Content	0.02% or less		JIS Z 3197
Cu Plate Corrosion Test	Pass		JIS Z 3197
Cu Mirror Corrosion Test	Pass		JIS Z 3197
Surface Insulation Resistance Test	Over 1.0×10 ⁹ Ω		JIS Z 3197 (40℃/90%RH, 168hrs)
Electro Chemical Migration Test	Over 1.0×10 ⁸ Ω No Migration		JIS Z 3197 (85℃/85%RH, 1000hrs)
Flux Content	11.5%	10.5%	JIS Z 3197
Viscosity	170Pa·s	190Pa·s	JIS Z 3284
Thixotropic Index	0.60	0.60	JIS Z 3284
Slump Test	Cold 0.2mm/Hot 0.3mm		JIS Z 3284
Tackiness Test	Over 0.9N within 24hours		JIS Z 3284
Solder Ball Test	Rank 2-3		JIS Z 3284
Validity	6 months		Unopened, Keep at 0-10℃

◇The values in this table are for reference.